PROVISIONAL PATENT DISCLOSURE

A Hybrid Local–Global Token-Mixing Architecture for Noise-Robust Image Classification Using Iterated Neural Cellular Automata

Draft for review — prepared to accelerate filing, not to replace patent-agent review

EVIDENCE COMMIT **5f8df70**	DATE OF DISCLOSURE **2026-08-12**
APPLICANT **[TO BE COMPLETED — institution/individual]**	INVENTOR(S) **[TO BE COMPLETED — mentor to confirm]**
FILING RECOMMENDATION **Provisional specification**	FIELD **Computer vision / neural network architecture**

4.3×
fewer parameters than the accuracy-matched baseline

+8.3pp
accuracy retained under additive sensor noise vs. best baseline

0%
overlap between architectures on ultrasound validation, 3 seeds

READ FIRST

What this document is: an engineering-authored technical disclosure, structured so a patent agent can draft the formal specification directly from §9. It is not itself a filing-ready legal document — see the two risk flags on the next page before sending to counsel.

§1 Before You Read Further

Two things need a patent agent's judgment before this goes anywhere. Everything else in this document is engineering fact, already verified against the codebase.

LEGAL RISK 1

Section 3(k) — algorithm exclusion. Indian patent law excludes "a mathematical method or a computer programme *per se*." Claim 1 (§9) is deliberately written as a system with a demonstrated technical effect (accuracy retained under sensor noise, §7), not as an abstract accuracy claim. Benchmark accuracy alone is exactly the kind of claim examiners reject under 3(k) — it is not the basis of any claim here.

LEGAL RISK 2

Section 3(i) — diagnostic method exclusion. Claim 5 involves medical ultrasound. It is worded as a system/apparatus and image-classification claim, not "a method of diagnosing a patient." **Have the patent agent confirm this framing survives 3(i) scrutiny before filing** — this is a legal call, not an engineering one.

ALREADY DECIDED

Provisional, not complete, specification. Chosen because active work (a data-efficiency sweep, additional seeds) would only strengthen the claims, not change them. Provisional secures a priority date now and gives 12 months to finalize. **Inventorship** is left as a placeholder on the cover page — settle who contributed to technical conception (architecture/experiment design) vs. supervision before the complete specification is due.

Every number in this document traces to a specific file and a command that reproduces it — see §10, Evidence Index, at the back. If a figure is ever questioned, that table says exactly where to look.

§2 Claims at a Glance

Six claims: one independent system claim, five dependent. Full legal wording is in §9 — this table is for a fast read.

#	COVERS	PLAIN-ENGLISH SUMMARY
1	**Independent** System claim	An image classifier with two stages: local iterated updates (fixed perception + zero-init update network + stochastic gating) followed by global self-attention — and it retains more accuracy than an attention-only system when the input has additive high-frequency noise.
2	Dependent	The local stage's perception uses three fixed (non-learned) filters: horizontal gradient, vertical gradient, identity.
3	Dependent	The local and global stages are roughly equal in size, local stage first — and this specific ordering is what produces the accuracy gain, not just "having both."
4	Dependent	The local stage's update network is zero-initialized, so it starts training as a pure identity mapping.
5	Dependent — applied embodiment	Applied to medical ultrasound: the "additive high-frequency noise" is **speckle**, ultrasound's dominant physical artifact — this is the claim carrying the §3(k) technical-effect argument.
6	Dependent	Classification pools patch tokens only; the network's classification token is never used, closing an obvious design-around.

§3 The Problem & Prior Art

The problem

- Vision Transformers classify images by comparing every patch against every other patch (self-attention) — expensive, and *learned*: the network must be taught from data that nearby pixels are related.
- When training data is limited, or images are noisy at test time, that learned locality is unreliable.

Closest prior art

WORK	WHAT IT DOES	GAP THIS DISCLOSURE FILLS
AdaNCA NeurIPS 2024 arXiv:2406.08298	Inserts a cellular-automaton module *between* attention layers. Attention stays primary throughout.	Attention is never replaced; robustness reported as one aggregate number, no breakdown by noise type, no mechanism.
ViTCA NeurIPS 2022 arXiv:2211.01233	Computes the cellular automaton's own update rule *using* self-attention.	Attention is fused in, not removed — a different architecture with a different cost profile.

NOVELTY GAP No prior work known to the applicant (a) fully replaces attention with iterated local processing in part of the network, (b) uses **fixed, non-learned** perception, or (c) demonstrates a **frequency-specific** robustness property — concentrated in one noise category, absent in others, with a working mechanism, rather than a general robustness claim.

§4 How the Invention Works

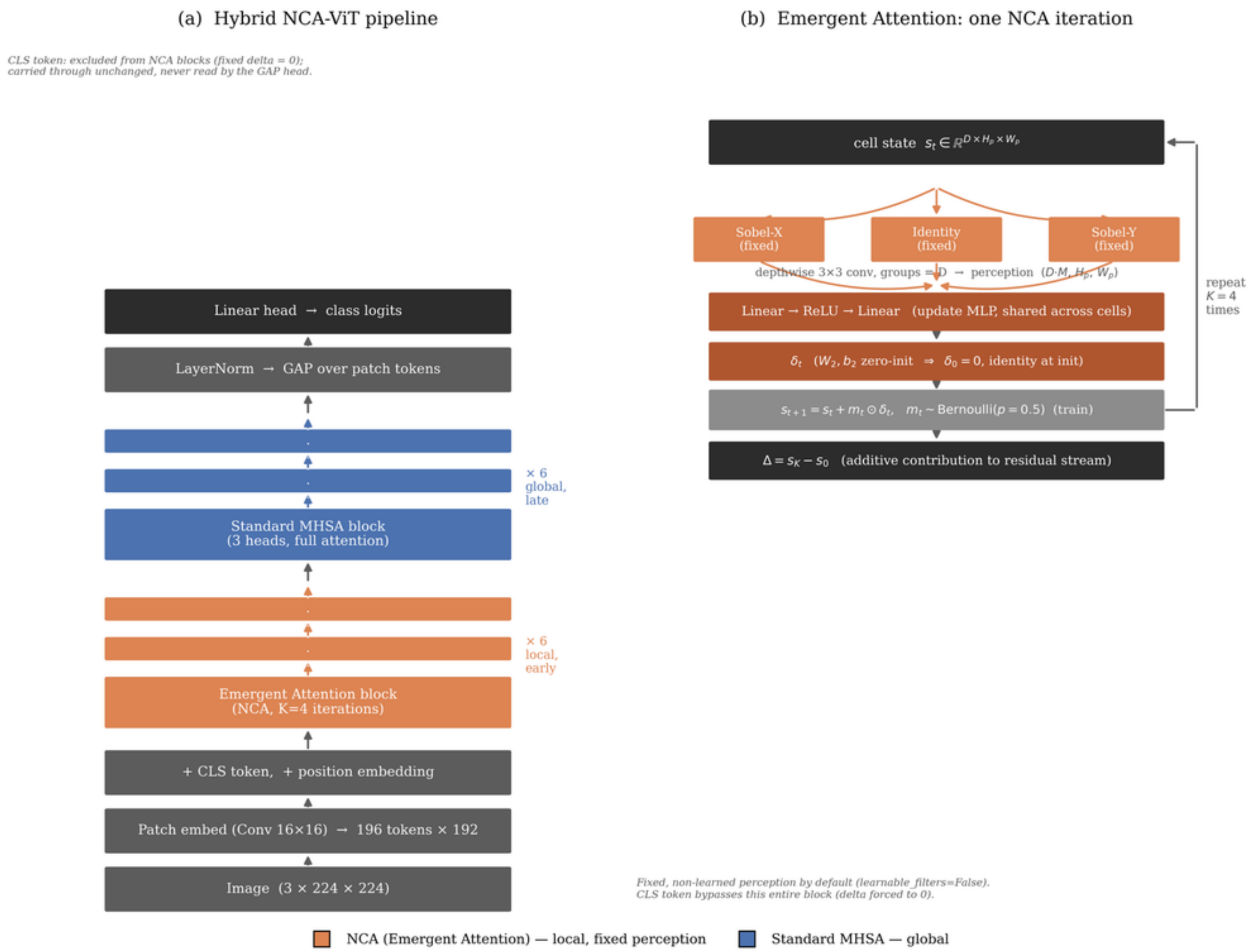

Fig. 1. (a) Full system: local stage (orange) → global stage (blue) → classifier. (b) One local-stage iteration.

Stage 1 — local, early (the novel part)

- Operates on a grid of image-patch tokens, repeated **K** times.
- Each iteration: (a) a **fixed** 3-filter perception step (horizontal/vertical gradient + identity) — no learned weights; (b) a small update network, **identical at every spatial location**, whose final layer starts at exactly zero; (c) the update is applied at each location only with some probability, sampled independently, during training only.
- The network's classification token is excluded entirely — it carries no image information through this stage.

Stage 2 — global, late

- Standard multi-head self-attention, unmodified, over the full token sequence. Positioned **after** Stage 1.

Classification

- The patch tokens (not the classification token) are pooled and passed to a linear layer.

Why the ordering matters — measured, not assumed

81.29–81.91%	76–77%	73.5–74%
accuracy at the 6:6 local:global split — the peak	accuracy with global stage only (0:12) — both neighbors score lower	accuracy with local stage only (12:0) — also lower

Five configurations tested (0:12 through 12:0), two independent random seeds. The 6:6 split beats both pure configurations at both seeds — this is what supports Claim 3.

§5 The Technical Effect (§3(k) Evidence)

This is the section that carries the patentability argument: a demonstrated effect on a stated technical problem, not an abstract accuracy improvement.

Result 1 — aggregate accuracy under corruption

Tested against four baseline architectures on 19 categories of image corruption (standard benchmark, Hendrycks & Dietterich, ICLR 2019). The invention scores **61.88%** mean accuracy vs. **59.37%** for the strongest attention-only baseline — which matches the invention's clean-image accuracy using **4.3× more parameters**.

Result 2 — the effect is category-specific, which is the actual evidence

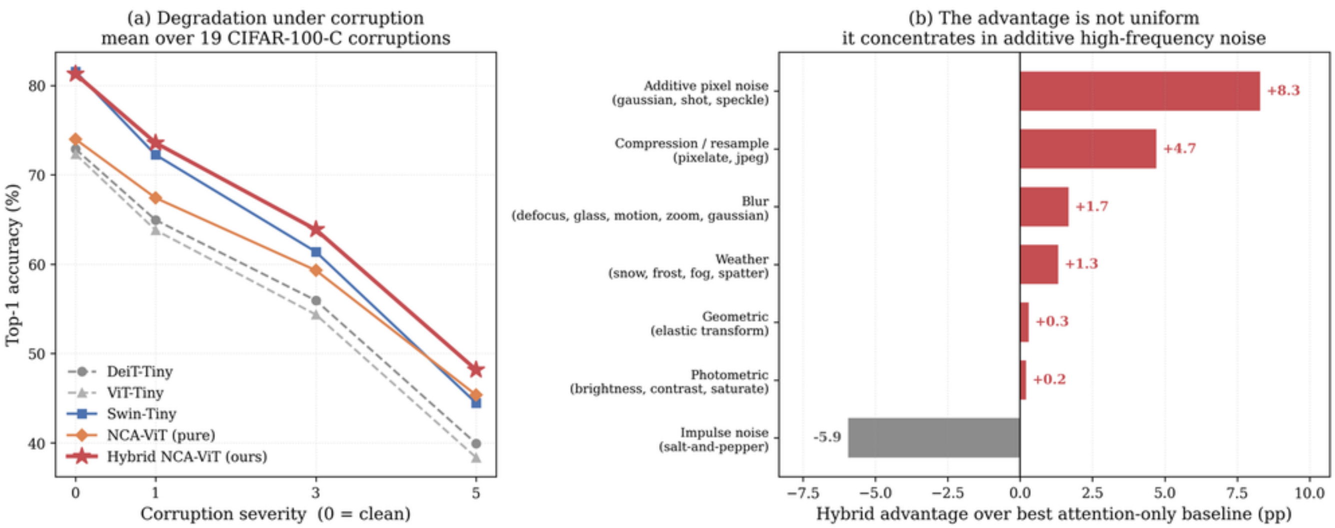

Fig. 2. Advantage over the best baseline, by corruption category — concentrated, not general.

Advantage is **+8.3 points** on additive high-frequency pixel noise, falling to near-zero on blur/weather/photometric, and **reversing to −5.9 points** on impulse (sparse-outlier) noise. A general-purpose accuracy improvement would not produce this shape — a specific physical mechanism does. This specificity is what distinguishes a technical effect from an algorithmic one.

Result 3 — mechanistic verification

Directly measured how far each architecture's internal representation moves under each corruption type. The invention's representation moves **16–26% less** than the baseline's on the corruption categories where it wins, and moves *more* — not less — on the one category (impulse noise) where it loses. This sign-correspondence between an internal measurement and the external accuracy result is direct evidence of a working mechanism, not an unexplained correlation.

§6 Applied Embodiment: Medical Ultrasound

Speckle — the single largest measured advantage (+9.6 points) — is also the dominant, physically unavoidable artifact of ultrasound image acquisition. Not an analogy: the same noise process.

Tested on BUSI, a public breast-ultrasound dataset (780 images, 3 diagnostic categories, 600 patients), trained from random initialization, against a size-matched baseline, at three independent random seeds:

49.83%
balanced accuracy, invention (3-seed mean)

36.39%
balanced accuracy, baseline (3-seed mean)

0 overlap
invention's worst seed beats baseline's best seed

Fig. 3. Balanced accuracy and macro-F1, invention vs. baseline, 3 seeds each.

This constitutes a demonstrated technical effect on a real, non-synthetic applied domain — satisfying the requirement that the invention solve a concrete technical problem in a stated field, not only improve an abstract benchmark metric.

§7 Formal Claims

FOR THE PATENT AGENT The claim language below is drafted for direct use in the specification. Mentor: you do not need to read this section word-for-word — §2 already gave you the plain-English summary. This is here so the agent doesn't have to draft claim language from scratch.

CLAIM 1 — INDEPENDENT (SYSTEM)

A computer-implemented image classification system comprising: **(a)** a patch embedding stage configured to divide an input image into a plurality of patches and produce a corresponding sequence of patch tokens, together with a designated classification token; **(b)** a first plurality of processing blocks configured to iteratively update the patch tokens over K iterations, wherein each iteration comprises: applying a fixed, non-learned spatial perception operation to the patch tokens arranged on a spatial grid; processing the resulting perception output through an update network shared identically across all spatial locations, said update network having an output layer initialized to zero-valued weights and biases; and applying the resulting update to the patch tokens, gated during a training phase by an independently-sampled random binary mask per spatial location and applied without gating during an inference phase; wherein the designated classification token does not participate in said iterative updating; **(c)** a second plurality of processing blocks, positioned after the first plurality of processing blocks in processing order, configured to perform multi-head self-attention across the full sequence of patch tokens and the classification token; and **(d)** a classification head configured to pool the patch tokens, following processing by the first and second pluralities of processing blocks, and to produce a classification output therefrom; wherein the system exhibits, relative to a comparable image classification system employing multi-head self-attention throughout without said first plurality of processing blocks, an increased retention of classification accuracy when the input image is degraded by additive high-frequency pixel-noise.

CLAIM 2 — DEPENDENT ON CLAIM 1

The system of Claim 1, wherein the fixed spatial perception operation comprises three depthwise-convolutional filters applied per channel: a horizontal spatial-gradient filter, a vertical spatial-gradient filter, and an identity filter, the weights of said filters remaining fixed and non-learned throughout a training procedure.

CLAIM 3 — DEPENDENT ON CLAIM 1

The system of Claim 1, wherein the first plurality of processing blocks and the second plurality of processing blocks are of substantially equal number, together constituting the full processing depth of the system, and wherein said substantially equal split with the first plurality preceding the second in processing order is configured to produce a classification-accuracy improvement, relative to processing orderings in which the proportion of the first plurality to the second plurality differs from substantially equal, or in which the second plurality precedes the first.

CLAIM 4 — DEPENDENT ON CLAIM 1

The system of Claim 1, wherein the update network comprises a first linear transformation, followed by a nonlinear activation function, followed by a second linear transformation, and wherein the zero-initialization of the second linear transformation's weights and bias causes each of the first plurality of processing blocks to perform an identity mapping at the commencement of a training procedure.

CLAIM 5 — DEPENDENT ON CLAIM 1 (APPLIED EMBODIMENT)

A method of classifying a medical ultrasound image using the system of Claim 1, wherein the additive high-frequency pixel-noise comprises speckle noise arising from coherent-wave ultrasound image acquisition, and wherein the system produces a diagnostic-category classification output with increased retention of classification accuracy relative to a comparable image classification system employing multi-head self-attention throughout, when applied to ultrasound images exhibiting said speckle noise.

CLAIM 6 — DEPENDENT ON CLAIM 1

The system of Claim 1, wherein the classification head is configured to pool the patch tokens by averaging, excluding the classification token, such that the classification output is independent of any representation carried by the classification token.

§8 Abstract (Filing Format, ~150 words)

An image classification system comprises a first plurality of processing blocks performing iterated, spatially-local updates using a fixed, non-learned perception operation and a zero-initialized, stochastically-gated update network, followed by a second plurality of processing blocks performing global multi-head self-attention. Ablation across five stage-ordering configurations at two independent random seeds confirms that positioning the local-update blocks before the attention blocks, at approximately equal depth split, maximizes classification accuracy. Evaluated against attention-only baseline architectures across nineteen categories of image corruption, the system demonstrates a classification-accuracy retention advantage concentrated specifically in additive high-frequency pixel-noise corruption, verified by direct measurement of internal representation stability. Applied to medical ultrasound image classification — where speckle noise, the corruption category showing the largest measured advantage, is the dominant acquisition artifact — the system outperforms a comparable self-attention baseline across three independent trained instances with non-overlapping performance ranges.

§9 Evidence Index

For verification by a patent agent or examiner. All paths relative to evidence commit `5f8df70`.

CLAIM / SECTION	EVIDENCE FILE(S)	REPRODUCE WITH
§4 stage-ordering table	`checkpoints/nca_vit_hybrid/cifar100/seed{42,123}/`	`python scripts/train.py model=nca_vit_hybrid model.nca_depth=N model.attn_depth=M data=cifar100`
§5 corruption results	`results/cifar100c_robustness.json`	`python scripts/evaluate_corruptions.py`
§5 mechanism / drift	`results/noise_drift.json`	`python scripts/measure_noise_drift.py`
§6 BUSI table	`results/busi_v2/, scripts/plot_busi_seeds.py`	`python scripts/train.py model=nca_vit_hybrid data=busi model.num_classes=3 training.augmentation.mixup_alpha=0 training.augmentation.cutmix_alpha=0 training.training.class_weighted=true training.training.seed={42,123,7}`
Architecture (§4, §9)	`src/models/hybrid_nca_vit.py, src/models/nca_attention.py`	`— (source code)`
Full corruption breakdown	`paper.md Appendix C`	`python scripts/evaluate_corruptions.py --full`

End of disclosure. Engineering-authored draft intended to accelerate, not substitute for, review by a registered patent agent — particularly the §3(k)/§3(i) framing flagged in §1.